Mining Neurodegenerative Genes Using Large Language Models

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Abstract

With the increasing popularity of a new artificial intelligence tool, **Large Language Models (LLMs)**, it is critical to understand what they can and can't do, especially in specialized professional settings like bioinformatics. This research underlines their strengths and weaknesses in recall and accuracy of genetic data mining and what this means for similar endeavors. It's also important to understand which prompting techniques and LLMs perform better than others, allowing us to further understand their characteristics. I tested three popular LLMs and six common prompting techniques to collect **F1 scores (a recall and accuracy confusion matrix metric)** for Alzheimer's disease, Parkinson's disease, and hypothetical neurodegenerative disease gene symbols. Each trial consists of prompting the LLM to return as many genes as possible for the respective disease and using the respective prompting technique. The data collected helps identify which prompting techniques and LLMs performed best for each disease. The results suggest that LLMs struggle with specialized data mining as they are noisy and fail to return enough data. So, it's important to be realistic with LLM performance expectations and focus on their strengths. However, all LLMs performed better when given examples and when generating genes for a hypothetical neurodegenerative disease. In comparison with the other trials, Google's Bard LLM found abnormal success in generating genes for hypothetical neurodegenerative diseases.

Methodology

- *Kyoto Encyclopedia of Genes and Genomes (KEGG)* [1] is the source of neurodegenerative disease gene symbols
- *HUGO Gene Nomenclature Committee (HGNC)*'s [2] Multi-Symbol Checker is used to translate KEGG symbols to HGNC-approved symbols for data consistency (some symbols in KEGG are aliases or otherwise not approved)
- I used *Python* (a succinct programming language) and *Google Colab* notebooks (documents to store, organize, and run code) to write programs to:
  1. Retrieve gene data over the internet from KEGG's application programming interface (API)
  2. Reformat and consolidate KEGG data into a list of HGNC-approved gene symbols
  3. Compute the F1 score for each trial by comparing the genes returned by the LLM and the HGNC ground truth
- I used consistent prompts that asked the LLM to return as many HGNC-approved genes as possible
- The prompt techniques I used are largely adopted from other sources and are some of the most common and well-known techniques today
- Brief explanations of each prompting technique used are as follows:
  1. Zero-shot: Provide zero examples before prompting [3]
  2. One-shot: Provide one example before prompting [3]
  3. Few-shot: Provide two or more examples before prompting (in this case, two) [3]
  4. Chain of thought: Provide a question and answer example framework before prompting [3]
  5. Tree of thoughts: Provide the prompt as a tree of steps that can be iteratively explored and evaluated by the LLM before arriving at an answer [3]
  6. Iterative refinement (3x): Prompting once as normal and then prompting twice more to refine the results (*note that this is slightly different from what is considered iterative refinement according to the source*) [4]
- I used three popular and accessible LLMs, Open AI GPT-3.5 [5], Open AI GPT-4 [5], and Google's Bard (using PaLM 2 as the underlying model) [6], as well as two well-known neurodegenerative diseases and one hypothetical
- I ignored duplicate genes to simplify the process, performed additional trials when the LLM failed to yield data, and began a new session for each trial to ensure the LLM did not have any prior context that could affect its response

Background

- There is a lack of knowledge surrounding LLMs and what they can be used for, even among its developers!
- LLMs have many limitations, including but not limited to:
  1. **Context, tokens (text), and memory** - only so much data can be retrieved at once
  2. **Training** - data not included in training generally cannot be retrieved, but there are exceptions
  3. **Lack of recency** - LLMs are generally only trained on data up to a certain date, but once again, there are exceptions
  4. **Accuracy** - LLMs are inherently noisy and generalized
- The results highlight how these limitations affect genetic data mining (and similar endeavors) and the interactions between prompting techniques, various LLMs, and the generation of different neurodegenerative diseases in maximizing the F1 score
- This is relevant to highlight LLM best practices and determine potential in specialized endeavors (that require substantial human time and effort, e.g., maintaining a genetic database!)
- Current literature involves some elements of this research, such as individual prompting techniques and LLM gene pathway extraction, but little in comparison among prompting techniques, for example

Research Questions

The following questions outline knowledge gaps this research provides insight and answers to:

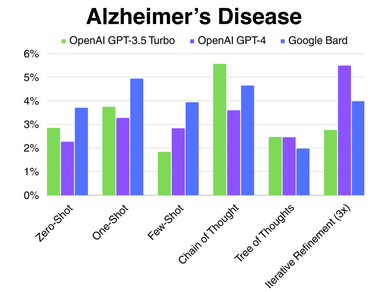
1. Can LLMs be a "source" of large specialized datasets such as neurodegenerative disease genes?
2. Can LLMs better come up with specific neurodegenerative disease genes or those of a hypothetical one?
3. What prompting techniques and LLMs are more effective at mining neurodegenerative disease genes?

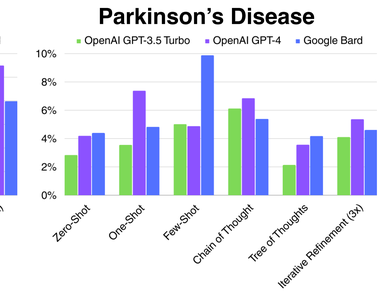
Note: As alluded to in the abstract, due to the ability of this research to be similarly replicated across disciplines and in various ways, it is reasonable to suggest that the results could apply generally. Accordingly, each of the questions can also be proposed generally. For example, what prompting techniques are (generally) most effective?

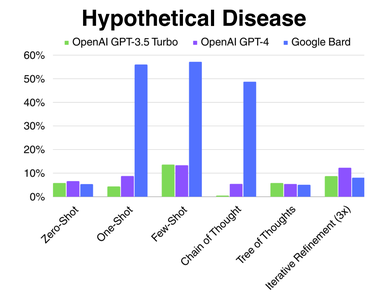
Challenges

1. Extensive time was spent in an attempt to devise a research topic. I attribute this to, ironically, the lack of knowledge of LLMs and what they can be used for, and the fact that I am a computer science undergraduate in a bioinformatics lab. Therefore, there are limitations to my knowledge in the biology domain.
2. Several pivots were made during the research process due to falling victim to the *Dunning-Kruger effect*:
   - I began by researching how to construct a knowledge graph on complex gene pathways from LLMs
   - I naively conjectured that this was too basic, so I proposed to research the impact of genetic and biological variations on cognitive abilities using LLMs
   - Realizing that I was wrong, I returned to the original research topic
   - Realizing that the original research was still considerably difficult, I landed on this research topic
3. The following are challenges I faced with researching the impact of genetic and biological variations on cognitive abilities:
   - It was difficult to find an accessible dataset relating biology to cognitive abilities
   - It was unclear what role LLMs would play in this research
4. The following and challenges I faced with the original research topic:
   - Comparing the quality of gene pathway graphs generated by LLMs is difficult because it requires a unique formula for computing recall and accuracy as there are not only genes involved but also interactions among them
   - LLMs generate text, and there is no standard format for creating a text-based graph
   - LLM length and accuracy challenges are exacerbated when generating text-based graphs based on datasets involving a few hundred genes
5. The following are challenges I faced with the current research topic:
   - Despite only prompting the LLM to return genes, there are still a few hundred genes involved in each disease which tests the length limits of LLMs and introduces inconsistencies among trials
   - LLMs don't respond in a consistent way, even when provided examples of how to respond
   - LLMs sometimes arbitrarily limit the amount of data returned, generally resulting in a lower F1 score for that trial

Data







Averages

Zero	One	Few	Chain	Tree	Iterative
4.23%	10.77%	12.51%	9.66%	3.68%	6.17%

GPT-3.5 Turbo	GPT-4	Bard (PaLM 2)
4.56%	5.79%	13.17%

Alzheimer's	Parkinson's	Hypothetical
3.47%	4.96%	15.09%

Bar graph details

- Each color represents one of three LLMs
- The x-axis displays the six prompting techniques used
- The y-axis represents the F1 score

Conclusion and Future Studies

- While promise is shown, it seems evident that LLMs struggle with data mining endeavors and cannot yet replace tasks or be a "source" of specialized datasets, facing length and accuracy challenges as expected
- Using the right LLM for the right task with the right prompting technique is key; LLMs are a tool, not a magic black box
- LLMs perform better at mining genes for a hypothetical neurodegenerative disease than an actual/specific one
  - This is likely due to factors such as an inherently higher fault tolerance and LLM generalization
- LLMs perform better when provided with examples on how to respond, with few-shot performing best overall
- Bard performed best, while GPT-3.5 Turbo performed worst
  - This is likely due to Bard being built for research and having access to real-time internet data, while on the other hand, GPT is the "industry standard" for natural language tasks [7]
  - GPT-4 outperformed GPT-3.5 likely because of its additional and improved features and ability to provide real-time data
  - All LLMs were "scared" to mine lots of genes in an attempt to increase accuracy, ironically decreasing its F1 score
  - However, Bard was least like this and more willing to generate an extensive list
- More "clever" prompting techniques faired worse than the more basic ones, suggesting more than just a diminishing return
- It seems sensible to leverage what was successful in this research to develop a plan for fine-tuning (customizing an LLM to enhance capabilities) and/or test the use of specific LLMs in *assisting* with specialized endeavors like this that an LLM may not be able to replace on its own

References

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Acknowledgements

This research project was facilitated by the University of Missouri College of Engineering, Digital Biology Lab, and Undergraduate Research Fellowship. Special thanks to Yibo Chen (Digital Biology Lab), Dong Xu (Digital Biology Lab), and the University of Missouri Undergraduate Research Office and S.T.A.R. (Student Training for Advancing Research) program.